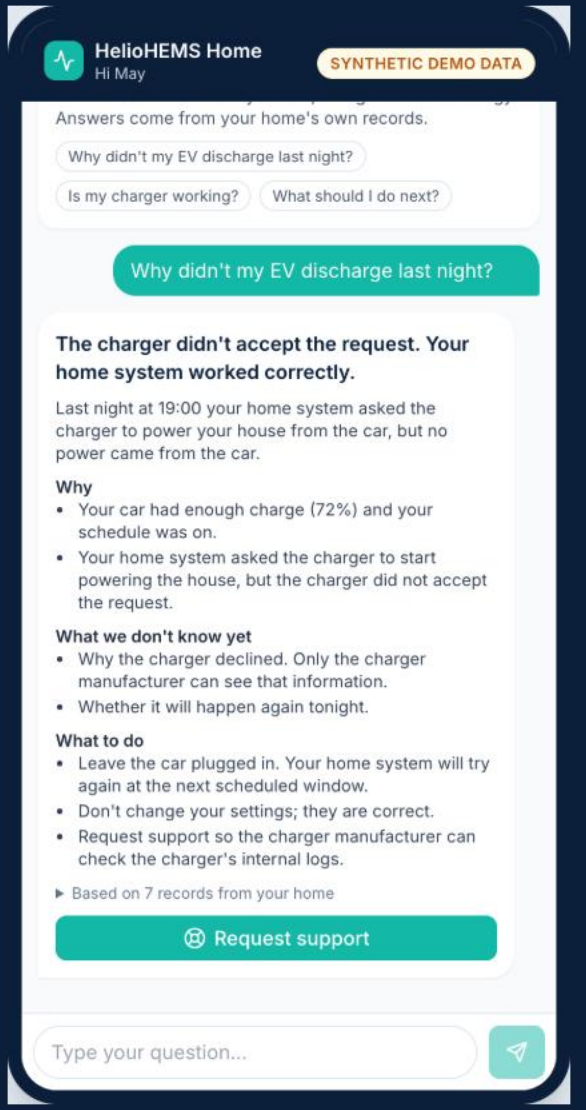
INVESTMENT-READY BUSINESS PROPOSAL

ConnectedCare AI

Evidence-grounded AI diagnostics for connected home energy, starting with vehicle-to-home and smart charging.

Phase 1 · B2B **Phase 2 · B2B2C** **Phase 3 · optional B2C**

10 November 2026 · Working title · Hypothesis-driven, pre-revenue business case



EXECUTIVE SUMMARY

A white-label support copilot that explains connected-energy failures from evidence

What

An AI diagnostics platform for HEMS and residential EV-charger manufacturers. V2H and smart charging are the first use cases.

How it grows

Phase 1 support copilot (B2B) → Phase 2 homeowner assistant in the OEM's app (B2B2C) → Phase 3 optional consumer product.

Value per OEM (base case)

€67k labour capacity a year, **2.1×** value/cost, **~5.7 months** payback. Illustrative, to be proven in a pilot.

Conditional GO

Fund customer discovery and a limited paid pilot, not a full launch. Ticket volumes, data access, willingness to pay and onboarding cost are not yet validated.

All financial figures are planning illustrations, not observed results.

1 · THE PROBLEM

One failed discharge leaves its evidence in five places, often from different vendors

HEMS
Setting
Schedule on, reserve 40%

→

HEMS
Command
“Discharge 3 kW” at 19:00

→

Charger cloud
Response
Session rejected, reason hidden

→

Meter / charger
Power
0.0 kW measured

→

CRM
Ticket
“My car didn’t power the house”

Today

Support reconstructs settings, commands, acknowledgements, power readings and errors by hand, across separate tools and IDs.

Result

Long investigations, weak escalations to partners, and homeowners repeating their story.

Example from the prototype’s Scenario B (synthetic data).

1 · JOBS TO BE DONE

The job: separate what is known from what is guessed, then hand over a support-ready case

> “When a customer reports a connected-energy failure, identify what is known, distinguish likely causes from missing evidence, propose approved next steps, and document a support-ready case.”

Fewer avoidable contacts

Less active investigation time

Better escalation evidence

Less repetition for homeowners

Guardrail: an instant AI response is not the same as a resolution. We measure resolution and investigation time, not reply speed.

2 · MARKET EVIDENCE

The signals are real, but none of them measures V2H support tickets yet

789,571

Fully electric cars, Netherlands, end-Sep 2026 (RVO)

Context for connected charging. Not the number of V2H homes or paying OEMs.

39 / 100

Problems per 100 home Level 2 chargers (J.D. Power, US, 2025)

Connectivity and app issues are a large category. A US survey, not a V2H ticket study.

~19M

Tickets in Freshworks' 2024 service benchmark

AI-copilot gains are for general service, not validated HEMS/V2H labour savings.

⚠ **Evidence gap:** no public baseline for technical tickets per HEMS installation, minutes per V2H incident or OEM willingness to pay. Next step: anonymised support exports and time-and-motion pilot tests.

Sources: RVO; J.D. Power 2025 US EV Home Charging Study; Freshworks 2024 Customer Service Benchmark. Links on the references slide.

3 · THREE-PHASE ROADMAP

Start inside the manufacturer, then reach the homeowner through the OEM's app

PHASE 1 · B2B · 0–12 MONTHS

Support copilot

Web console for L1/L2 support at a HEMS or charger OEM. Connects CRM, installation registry, commands, telemetry and manuals.

Goal: paid pilot and measured agent time saved.

PHASE 2 · B2B2C · 12–24 MONTHS

Homeowner assistant

White-label assistant inside the OEM's mobile app. Plain-language answers, safe checks and consent-based handoff to the same case workflow.

Goal: measured deflection, per-home revenue.

PHASE 3 · B2C · 24+ MONTHS · OPTION

Multi-vendor assistant

Independent homeowner product using authorised partner APIs and local gateways. A new business model, not automatic expansion.

Gate: consumer willingness to pay and acquisition cost.

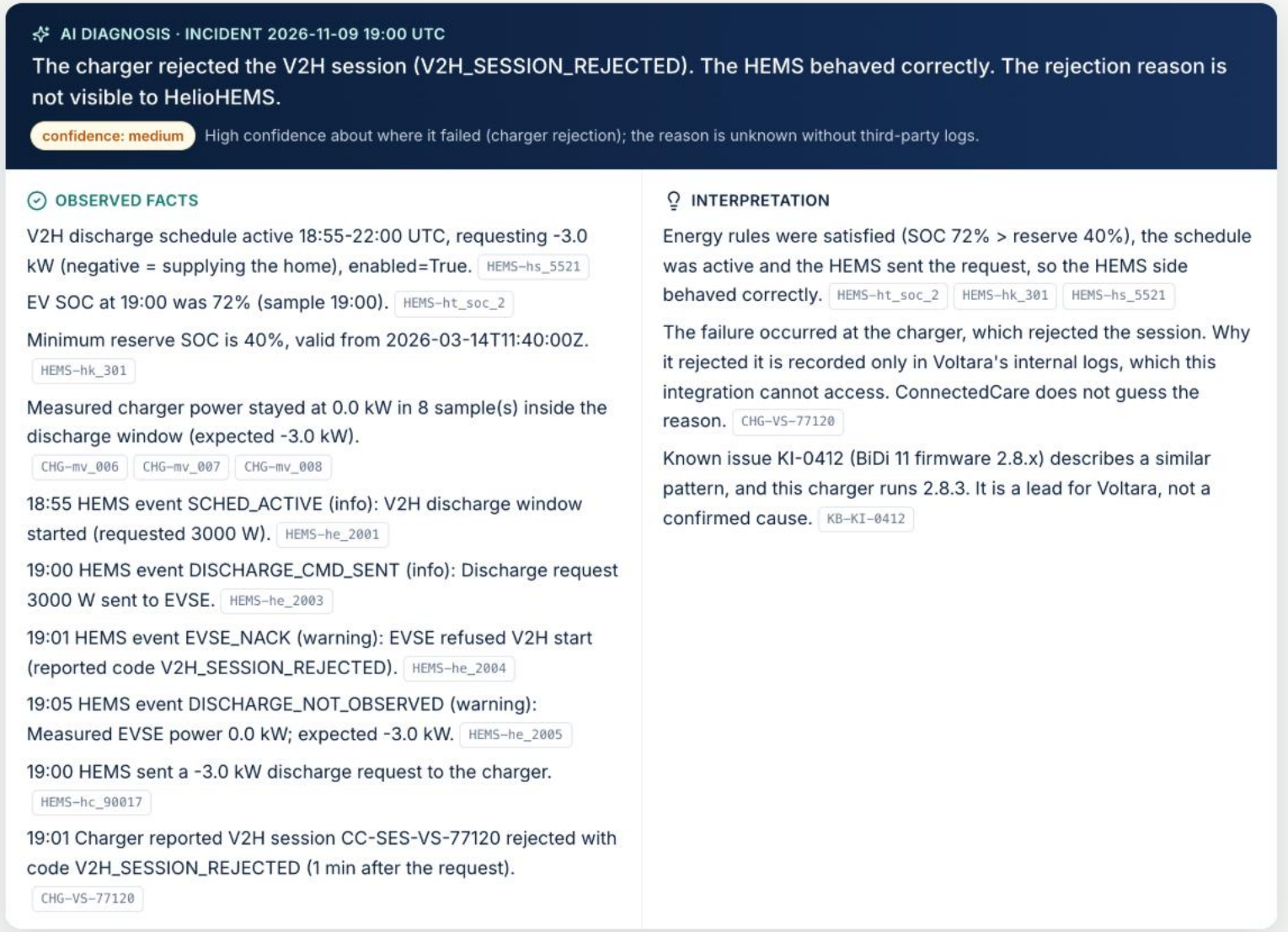
Each phase starts only if the previous stage gate proves economic value, accuracy and repeatable integration.

PHASE 1 · SUPPORT COPILOT (B2B)

Support staff get a diagnosis with its evidence attached

- Case queue with CRM customer, installation and verified devices
- Observed facts, interpretation, missing information and actions, each linked to source records
- Confidence stated, unknowns disclosed
- Escalation drafted for the right partner

Buyer: Head of Support · Users: L1/L2 agents

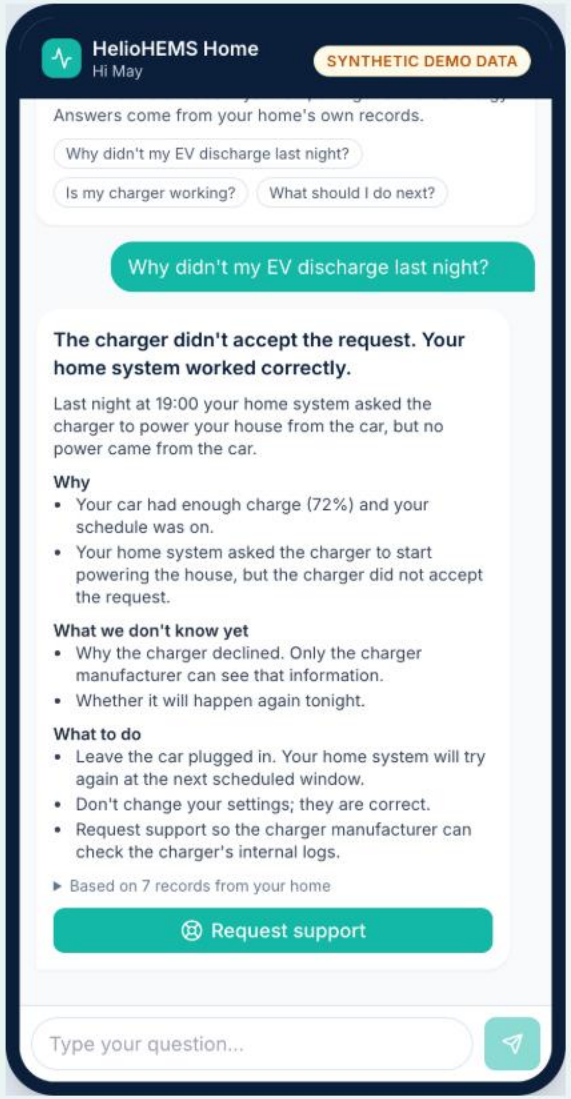


PHASE 2 · HOMEOWNER ASSISTANT (B2B2C)

Homeowners get a plain answer, and share evidence only with consent

- “Why didn’t my EV discharge?” answered from the home’s own records
- Says what is unknown and who can find out
- Shows exactly which records would be shared before the homeowner agrees
- Creates a support case with the evidence already attached

White-label, inside the OEM’s iOS / Android app or PWA





END-TO-END JOURNEY

From the homeowner's question to a partner escalation, with no re-typing

1 · Ask
Homeowner asks in the app

→

2 · Check
Agents read schedule, command, charger status, power

→

3 · Explain
Plain answer that names the unknowns

→

4 · Consent
Homeowner approves what is shared

→

5 · Resolve
Support continues with full evidence, escalates if needed

1 case
Homeowner app and support console share one record

3 views
Homeowner, support and partner each see only what they may

0 actions
on equipment without a human. The MVP is read-only

Demonstrated end to end in the prototype across three scenarios (synthetic data).

HOW IT WORKS · AGENTS

One orchestrator, four specialist agents, fourteen deterministic tools

Orchestrator

Claude (LLM) picks agents and writes the answer. Offline rules as fallback.

→

Technical Diagnostics

Was the command sent, accepted, rejected? Charger status, errors.

4 tools · devices, commands, sessions, events

Energy Behaviour

Battery vs reserve, schedules, expected vs measured power.

4 tools · schedules, measurements, constraints

Security Correlation

Setting changes near the incident. Flags for review, never declares an attack.

2 tools · configuration change log

Support Resolution

CRM history, manuals, known issues, escalation context.

3 tools · customers, past cases, knowledge base

Agents are read-only and see only the unified data model. One shared tool looks up any evidence record by ID.

HOW IT WORKS · AGENTS IN THE PRODUCT

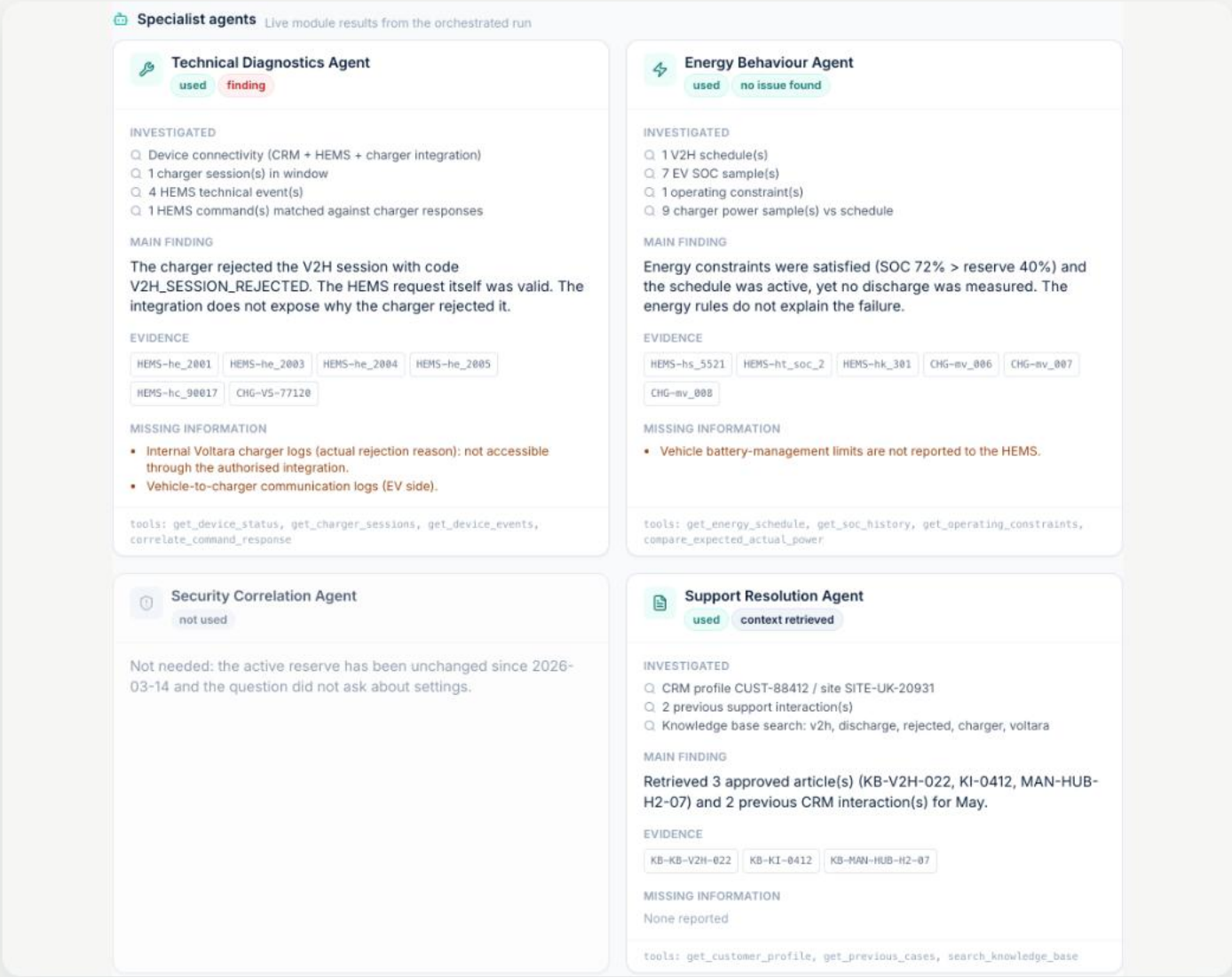
Every agent shows its work

What it checked
Records and tools used, in plain words

What it found
One finding, linked to evidence IDs

What it couldn't see
Missing data is declared, not guessed

When it wasn't needed
Marked "not used", with the reason



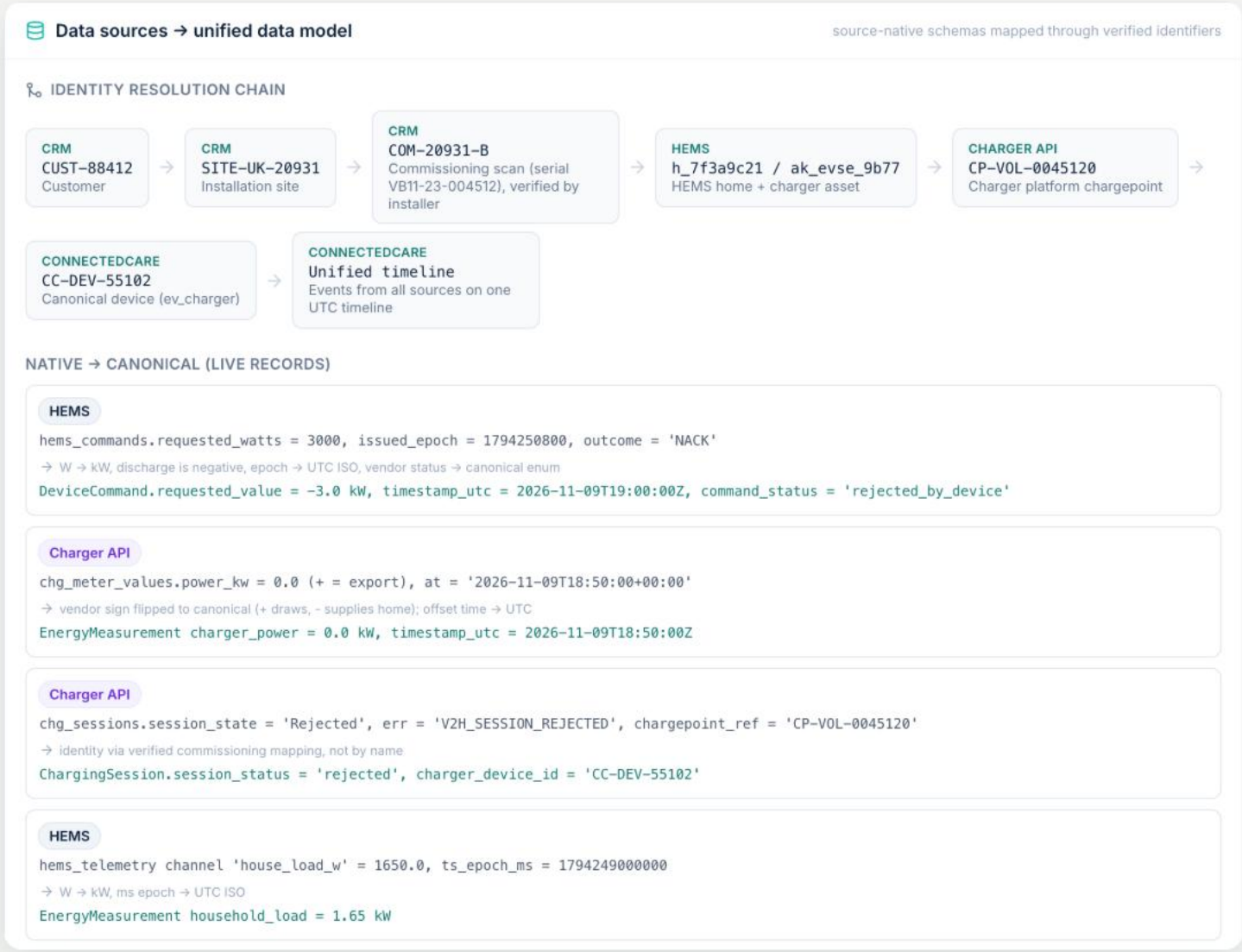
HOW IT WORKS · UNIFIED DATA MODEL

Four sources, one verified view of the home

- **CRM / ticketing**
- **HEMS backend**
- **Charger partner API**
- **Manuals + audit log**

Adapters → canonical model

- Units, signs and time zones normalised
- Devices matched by commissioning scan and serial, never by nickname
- Every record keeps its source and ID
- Partial visibility is explicit: no access to other vendors' internals



HOW IT WORKS · GOVERNANCE

The LLM explains; deterministic tools decide what is true

LLM does

- Understand the question
- Choose which agents to run
- Write role-appropriate language

Tools do

- Retrieve data and calculate
- Resolve device identity
- Enforce permissions

Humans approve

- Sensitive case updates
- External escalation
- Anything touching equipment

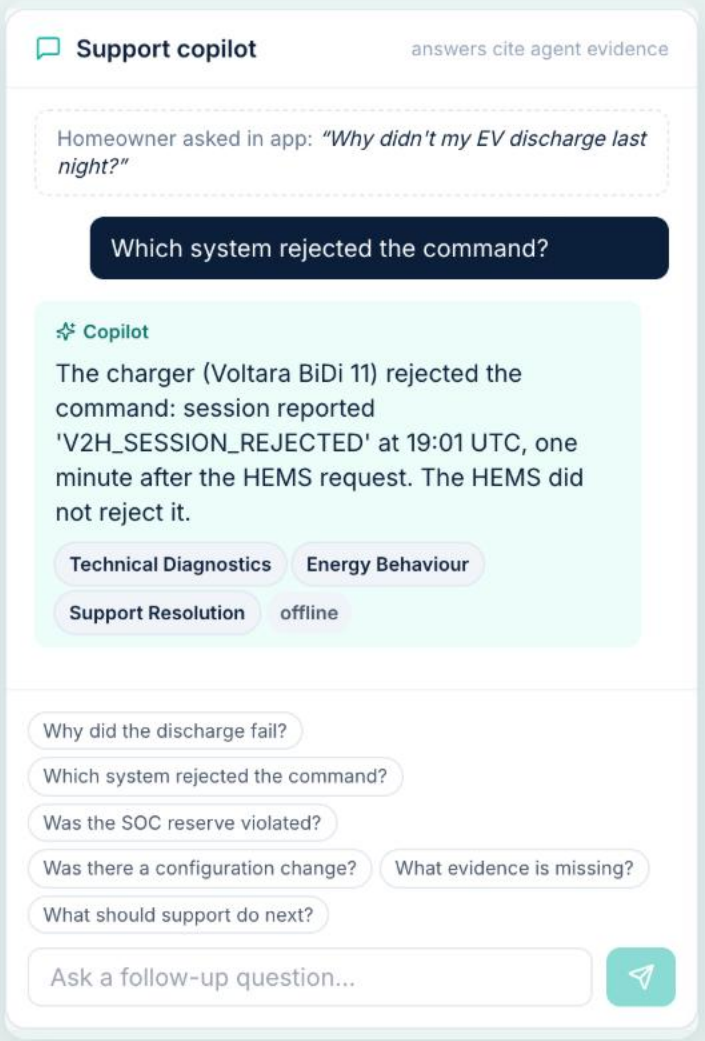
Visibility tier	Homeowner	Support	Charger partner	Example
Homeowner-safe	Yes	Yes	Yes	Schedule, battery %
Support-internal	No	Yes	Yes	Error codes, sessions
Restricted	No	Yes	No	Security audit log

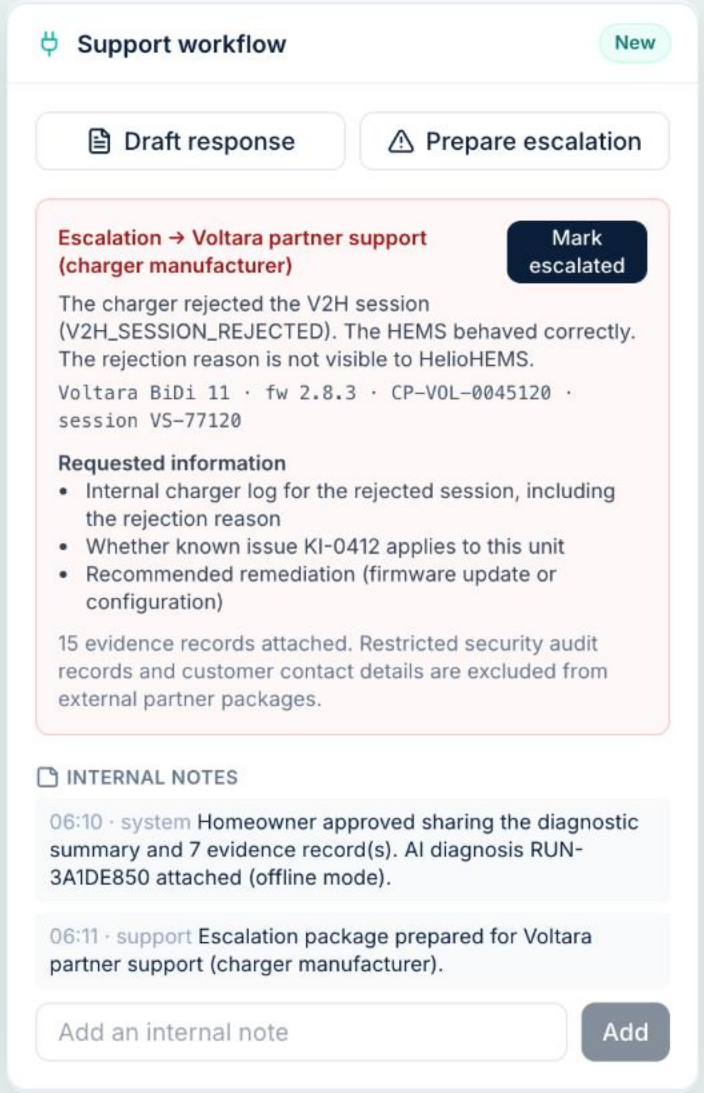
Answers may cite only evidence an agent returned; a homeowner answer that leaks internal detail is rejected. Missing records are declared, never invented.

SUPPORT WORKFLOW

Ask follow-ups, then escalate with a ready package

- Copilot answers follow-up questions and names the agents it used
- Escalation goes to the right partner with session IDs, firmware and requested information
- Restricted security records and contact details are left out for external partners
- Draft reply to the homeowner uses only homeowner-safe facts





4 · SEGMENTATION AND BUYING CENTRE

Target OEMs with large fleets, recurring tickets and accessible telemetry

Segments

Initial target

HEMS software providers and residential smart-charger OEMs

Adjacent

Battery and inverter makers, energy-service platforms, installers, aggregators

Qualify on

Ticket volume, issue mix, support cost, data access, willingness to co-fund

Buying centre

Role	Who
Economic buyer	Head of Support / Customer Operations
Champions	L1/L2 technical support, after-sales
Gatekeepers	CTO, engineering, security/privacy, procurement
Phase 1 user	Manufacturer support employee
Phase 2 user	Homeowner, via the OEM app

STP: segment by OEM type, fleet and data readiness · target high-volume support teams · position as diagnostics, not a chatbot.

8 · COMPETITIVE ANALYSIS

Incumbents own the workflow or the device; the gap is cross-source diagnosis

Alternative	Workflow	Telemetry	Cross-vendor	Energy rules	Handoff
Zendesk AI / Freshworks Freddy	Strong	Limited	Limited	Limited	Partial
OEM internal diagnostic portal	Limited	Own only	Limited	Strong	Limited
Charger-management platform	Limited	Own only	Limited	Partial	Limited
Home Assistant	None	Partial	Partial	Partial	None
Custom LLM + RAG (build)	Partial	Partial	Partial	Partial	Partial
ConnectedCare (target)	Integrates	Via OEM APIs	**Strong**	**Strong**	**Strong**

⚠ **Build-versus-buy is the main threat.** Defensibility must come from reusable connectors, validated diagnostic cases, quality benchmarks and lower deployment cost. No exclusive advantage is established yet.

Ratings are a team assessment from public positioning, to be validated in discovery.

9 · STRATEGY ANALYSIS · SWOT

Domain focus is the strength; unproven economics is the weakness

Strengths

- Energy-domain specificity
- Shared evidence across homeowner and support
- Working multi-agent prototype

Weaknesses

- Support economics unproven
- Integrations not yet built with a real OEM
- Small team, no references

Opportunities

- Connected-energy growth (EVs, V2H, HEMS)
- White-label distribution through OEM apps
- EU Data Act improves data access

Threats

- Incumbents and OEM in-house builds
- Closed vendor APIs
- AI errors; slow enterprise procurement

Implication: lead with a paid pilot that proves accuracy and time saved before scaling integrations.

9 · STRATEGY ANALYSIS · PORTER'S FIVE FORCES

An attractive problem in a tough market structure

High

Rivalry

Generic AI support suites and device platforms already sell to OEMs

High

Buyer power

Few large OEMs, long procurement, they control the data

High

Substitutes

In-house LLM + RAG, more support staff, better manuals

Medium–high

New entrants

LLM tooling is cheap; validated energy connectors and cases are not

Medium

Supplier power

LLM providers are interchangeable; device-data owners are not

So what

Win on proof and cost

Benchmarked accuracy, fast onboarding, reusable connectors

Ratings are a qualitative team assessment.

9 · STRATEGY ANALYSIS · PESTEL

Tailwinds from the energy transition and data rights; privacy is the constraint

Political

EU push for electrification and grid flexibility favours V2H and smart charging

Economic

Support labour is costly; OEMs under margin pressure look for efficiency

Social

Homeowners expect app-based, instant answers but distrust opaque AI

Technological

LLM tool-calling is mature; OCPP 2.0.1 and partner APIs standardise device data

Environmental

Reliable V2H keeps more home energy local and supports flexibility

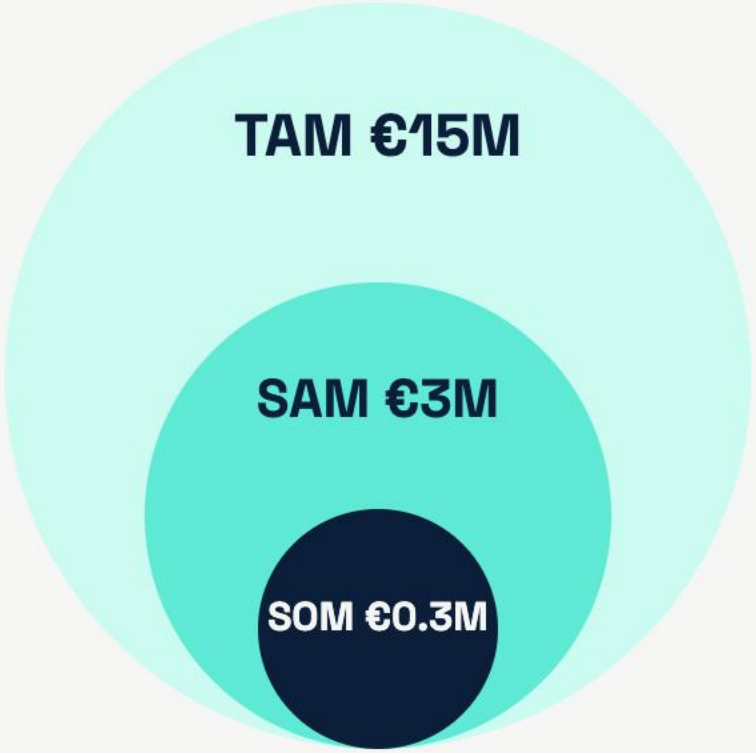
Legal

EU Data Act on connected-product data; GDPR consent and privacy duties

Data-sharing permission and technical interoperability are separate questions: never assume one OEM can read another vendor’s cloud.

10 · MARKET SIZING (ILLUSTRATIVE)

Bottom-up: paying OEM accounts × contract value, not EVs or homes



Level	Assumption	Annual value
TAM	500 eligible European organisations × €30k	€15,000,000
SAM	100 reachable accounts × €30k	€3,000,000
SOM	10 acquired accounts × €30k	€300,000 ARR

Buyer counts and contract value are unverified placeholders. A narrow V2H niche alone may not support venture-scale returns: growth needs more device categories, countries and modules.

Next step: a sourced census of target accounts to replace the placeholders.

11 · PRICING AND BUSINESS MODEL

Paid pilot first, then annual SaaS with a per-home add-on in Phase 2

Paid pilot, 6–8 weeks	€5k–€15k
Support copilot / year	**€18k–€30k**
Onboarding (one-off)	€3k–€15k
Phase 2, per active home / month	€0.10–€0.40

Customers
HEMS and charger OEMs

Value
Less effort, better triage and escalations

Channels
Founder-led sales, design partners

Key resources
Connectors, canonical model, case benchmark

Partners
OEMs, CRM vendors, installers

Costs
Integration, LLM and hosting, security

Pricing figures are discovery hypotheses, not market quotes.

12 · CUSTOMER SAVINGS

One OEM frees €16k–€200k of support capacity a year, depending on volume

Capacity value = annual tickets × AI-assisted share × minutes saved ÷ 60 × loaded labour €/hour



Gross labour capacity, not realised cash savings. Bars to scale. All inputs are illustrative until measured in a pilot.

12 · BASE-CASE ECONOMICS FOR THE CUSTOMER

Base case returns 2.1× its cost and pays back in under six months

Capacity value	€67,200
Annual licence	– €24,000
Ongoing overhead	– €8,000
Net capacity value	**€35,200**

Initial onboarding excluded.

2.1×
value / cost

~110%
net return on cost

5.7 mo
payback, if value accrues evenly

4,000
AI-assisted tickets to break even (~5,715 total at 70%)

Phase 2 add-on: 20,000 homes × 0.30 tickets/home/year × 20% truly deflected × 20 min × €40/h = **€16,000** more capacity, without double-counting Phase 1 cases.

Cost basis: €24k licence + €8k overhead = €32k/year. Break-even at 12 min saved × €40/h = €8 per assisted ticket.

13 · STARTUP UNIT ECONOMICS

About 21 full-year customers cover €500k of fixed costs

€300k
ARR at 10 customers × €30k

21
customers to break even: €500k ÷ (€30k × 80%)

€900k
ARR at 30 customers × €30k

Margin target 75–85%

Needs controlled LLM, hosting, support and recurring integration costs. Custom integrations can destroy an otherwise good SaaS margin.

KPIs we track

Sales cycle · pilot-to-paid conversion · onboarding hours · cost to serve · gross margin · churn · CAC · net revenue retention · ACV

Sensitivity calculation before tax and financing, not a forecast.

GROWTH · ANSOFF MATRIX

Four ways to grow from the same data model, from safest to boldest

	Existing product (diagnostics copilot)	New products and modules
Existing markets	**1 · Market penetration · low risk** More HEMS and charger OEMs; widen from V2H to all smart-charging and HEMS tickets	**2 · Product development · medium** New modules on the same model: proactive fault alerts, fleet analytics, warranty/RMA insight, installer quality checks
New markets	**3 · Market development · medium** New energy segments (batteries, heat pumps, retailers, aggregators) and new countries; B2B2C via OEM apps	**4 · Diversification · high risk** Independent B2C multi-vendor home-energy assistant

Sequence: 1 → 2 and 3 in parallel → 4 only after the stage gates. The unified data model is what makes 2 and 3 cheaper than starting from scratch.

GROWTH · GE–MCKINSEY SEGMENT MATRIX

Win chargers and batteries first; enter retailers and aggregators selectively

High	Heat-pump makers	**Energy retailers / platforms**	**Charger OEMs** **Battery / inverter makers**
Med		Aggregators / VPP	**HEMS software providers**
Low			Installers (channel / light tier)

Vertical: market attractiveness: account size, ticket pain, willingness to pay.

Horizontal: our ability to win, from low (left) to high (right): data-model fit, data access, existing proof.

Invest / grow

Selective growth

Later or as partner channel

Placement is a qualitative team assessment, to be scored with discovery data. Ability to win rises left → right.

GROWTH · THE DATA MODEL AS AN ASSET

Each new segment reuses the core model and adds only a few entities

Segment	Core reused	New entities needed	New module
HEMS + charger OEMs	All 15 entities	None (today's model)	Diagnostics copilot
Battery / inverter makers	All core	Battery state, inverter mode, grid limit	Fault alerts
Heat-pump makers	All core	Thermal measurement, comfort setpoint	Comfort diagnostics
Energy retailers / platforms	Customer, device, case, consent	Tariff, contract, billing event	Bill-shock explainer
Aggregators / VPP	Device, command, measurement	Flex event, dispatch, baseline	Dispatch audit
Installers	Installation, device, mapping	Commissioning checklist	Install QA

More connectors → **More validated cases** → **Faster, cheaper onboarding** → **Higher margin, more customers**

New entities are design proposals. Any cross-customer benchmark data needs contracts, consent and anonymisation.

GROWTH · B2B REVENUE BUILD-UP (ILLUSTRATIVE)

Adding energy segments grows the B2B pool from €4.2M to about €20M

Segment (Europe)	Accounts	ACV	TAM	Y5 clients	Y5 ARR
HEMS software	60	€30k	€1.8M	8	€240k
Charger OEMs	80	€30k	€2.4M	10	€300k
Battery / inverter makers	120	€35k	€4.2M	8	€280k
Heat-pump makers	60	€40k	€2.4M	3	€120k
Energy retailers / platforms	100	€50k	€5.0M	5	€250k
Aggregators / VPP	40	€60k	€2.4M	3	€180k
Installers (light tier)	300	€6k	€1.8M	20	€120k
Total	**760**	—	**€20.0M**	**57**	**€1.49M**

Modules on the data model (alerts, analytics, install QA): ~30% of 37 enterprise clients × €16k ≈ **€0.18M** more ARR. Core today (HEMS + chargers) = €4.2M TAM.

Account counts and ACVs are unverified planning assumptions; replace with a sourced account census. Y5 = year 5 base case.

GROWTH · B2B2C REVENUE (ILLUSTRATIVE)

Revenue scales with active homes in partners' apps, not with sales headcount

Annual revenue = active homes × €/home/month × 12

Active homes	€0.10	€0.25	€0.40
100,000	€0.12M	€0.30M	€0.48M
250,000 (Y5 base)	€0.30M	**€0.75M**	€1.20M
500,000	€0.60M	€1.50M	€2.40M
1,000,000	€1.20M	€3.00M	€4.80M

Y5 base case

20 partners with apps × 50k connected homes = 1M homes; 25% opt in → 250k active × €0.25

Why partners pay

Deflected contacts, better app ratings, consented evidence when cases do reach support

Price range from the business plan (€0.10–€0.40). Partner count, homes per partner and opt-in are assumptions.

GROWTH · B2C OPTION · LTV / CAC

B2C works only if a homeowner costs under about €40 to acquire

€48
a year (€4/month)

×

75%
gross margin

×

3.3 yrs
lifetime (30% churn)

=

~€120
LTV → CAC ≤ €40 for 3:1

Subscribers	Annual revenue
10,000	€0.48M
25,000 (upside case)	€1.20M
50,000	€2.40M

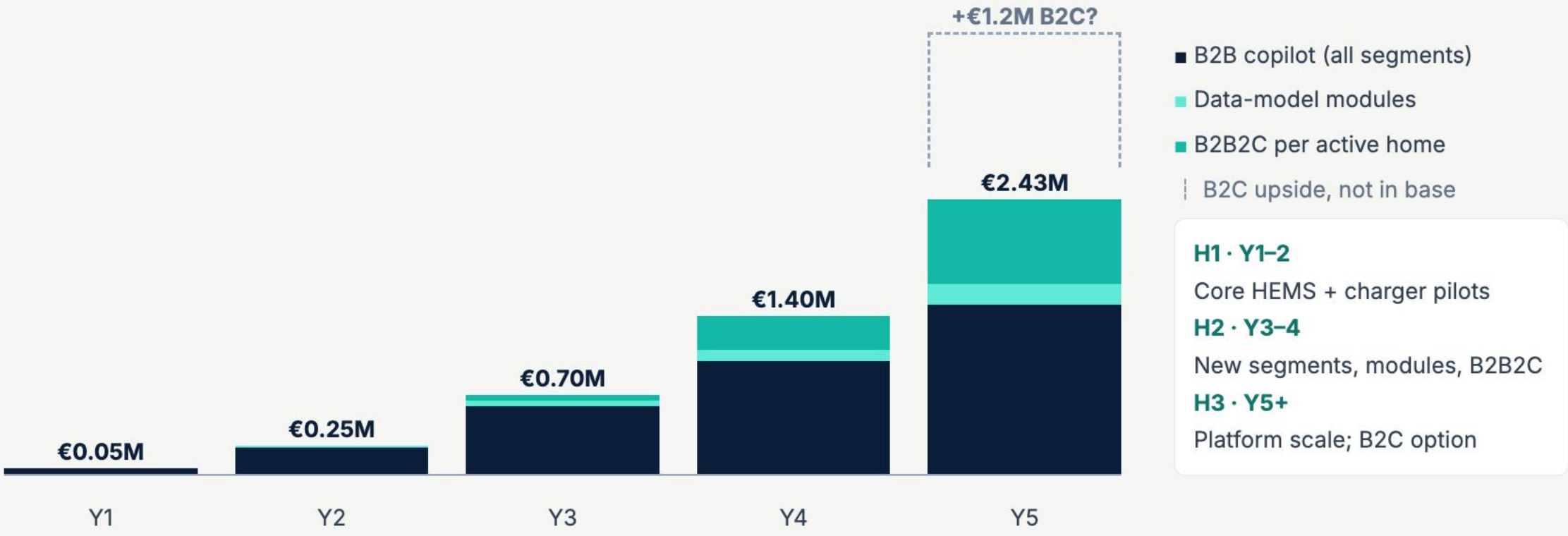
What you must believe

- Authorised multi-vendor data access
- A consumer brand and app-store channel
- No conflict with OEM partners (channel risk)

Price, margin and churn are assumptions for illustration. B2C is not in the base case.

GROWTH · THREE HORIZONS, 5-YEAR REVENUE (BASE CASE)

From pilots to about €2.4M ARR by year 5, with B2C as upside



Bars to scale (120 px per €1M). Y5 = €1.49M B2B + €0.18M modules + €0.75M B2B2C. Illustrative plan, not a forecast.

GROWTH · BUSINESS MODEL COMPARISON

B2B builds the asset, B2B2C scales it, B2C is a separate bet

	B2B	B2B2C	B2C (option)
Revenue driver	Accounts × €6k–€60k ACV	Active homes × €0.10–€0.40/month	Subscribers × ~€4/month
Pool	~€20M TAM, 760 accounts	€3M per 1M active homes at €0.25	Unsized; needs consumer research
Y5 base	€1.67M incl. modules	€0.75M	€0 (upside €1.2M)
Margin	High once connectors reuse	High; LLM cost per home to watch	Lower: acquisition and support
Capital need	Low–medium	Low (rides on B2B)	High (marketing, brand)
Key dependency	OEM data access	Partner app integration	Multi-vendor APIs, CAC ≤ €40
Main risk	Build vs buy	Low opt-in	Channel conflict with OEMs

Recommendation: fund B2B across energy segments, attach B2B2C to every OEM contract, keep B2C as a real option to test in year 3+.

All figures are illustrative planning assumptions built on the business plan's pricing hypotheses.

14 · RISKS AND MITIGATIONS

Every critical risk has a cheap test before we scale

Risk	Severity	Mitigation
Low ticket volume per OEM	High	Qualify prospects on ticket data before any pilot
Limited vendor telemetry	High	Design for partial visibility; state what is unknown
No uplift over incumbents	High	Blinded test on historical cases vs current tools
Hallucinated technical advice	High	Deterministic checks, cited evidence, human approval
Integration cost	Medium	Standard connectors and canonical model
Privacy / security failure	Medium	Role-based access, consent, audit logs, tenant isolation
Slow procurement	Medium	Small paid pilots with a named budget owner

Phase 3 risk (consumer willingness to pay) is tested only if Phases 1–2 succeed.

14 · VALIDATION PLAN

The pilot passes only if it beats today's workflow on time and accuracy

10–15
OEM stakeholder interviews

2
prospective design partners

50–100
anonymised historical cases

Blinded
comparison with the current workflow

Proposed acceptance targets

≥20%
less active investigation time

≥90%
diagnostic accuracy on answerable cases

≥95%
evidence traceability

0
critical fabricated diagnoses

≥1
paid conversion

Targets are proposed, not observed.

15 · GO-TO-MARKET AND MILESTONES

From interviews to 2–3 paying OEMs in twelve months, if the evidence holds

Months 0–2	Months 2–4	Months 4–6	Months 6–12	Months 12–24	Months 24+
Interviews, buyer map, data audit, recruit design partners	MVP on synthetic data (done) and evaluation suite	Paid OEM pilot	2–3 paying OEMs if pilot passes	White-label homeowner assistant, deflection measured	Decide on independent B2C

Stage gate: no expansion unless the pilot proves economic value, technical accuracy and repeatable integration.

Channel: founder-led direct sales and paid design partnerships.

16 · INVESTMENT DECISION

Fund the proof, not the launch

Conditional GO

Customer discovery and one limited paid pilot with an OEM design partner.

NO-GO for now

Full launch, until ticket economics, paid buyer interest, data access and differentiation are validated.

Why not generic Zendesk AI?

Device-aware diagnosis across sources, with evidence. Must be shown in the pilot.

Why will OEMs share data?

They pay us to use their own authorised systems. No presumed third-party access.

Why V2H first?

A compelling technical wedge; adjacent charging and HEMS issues bring the volume.

REFERENCES AND EVIDENCE STATUS

Sources

- RVO, electric mobility statistics (Netherlands)
- J.D. Power, 2025 US EV Home Charging Study
- Freshworks, 2024 Customer Service Benchmark
- Zendesk AI overview
- Home Assistant energy documentation
- European Commission, Data Act explained
- Open Charge Alliance, OCPP 2.0.1

Quantitative scenarios, prices, customer counts and targets are illustrative, not observed performance. Product screenshots come from the ConnectedCare prototype running on synthetic data for the fictional OEM HelioHEMS.